

EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 5

This homework is due Friday, October 2, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Proving DTFS Properties

Let $x[n]$ be a p -periodic discrete-time signal with DTFS coefficients $\{X_k \mid k \in \langle p \rangle\}$, where $\langle p \rangle$ is any discrete interval of p contiguous integers.

For example, $\sum_{k \in \langle p \rangle} X_k$ may denote $\sum_{k=0}^{p-1} X_k$, or $\sum_{k=-p/2}^{p/2-1} X_k$ for even p . Throughout, $\omega_0 = \frac{2\pi}{p}$.

(a) Let $k_0 \in \mathbb{Z}$, and define the modulated signal

$$y[n] = e^{ik_0 \omega_0 n} x[n].$$

Let Y_k denote the DTFS coefficients of $y[n]$. Prove that

$$Y_k = X_{k-k_0}, \quad \forall k \in \langle p \rangle.$$

This is called the **Modulation** property. It is the ‘dual’ of the **Time Shifting** property. Think about how ‘dual’ is an appropriate adjective here, intuitively and mathematically.

Hint: try writing down the analysis equation for both of Y_k and X_{k-k_0} .

(b) Now suppose $x[n]$ is real-valued and odd, i.e. $x[n] = -x[-n]$ for all $n \in \mathbb{Z}$. Show that X_k is odd in k and purely imaginary. More formally, show that

$$X_{-k} = -X_k \quad \text{and} \quad X_k^* = -X_k, \quad \forall k \in \langle p \rangle.$$

Hint: for the first claim, try a change of variables $m = -n$ in the analysis equation for X_{-k} . For the second, recall that a complex number z is purely imaginary if and only if $z^ = -z$.*

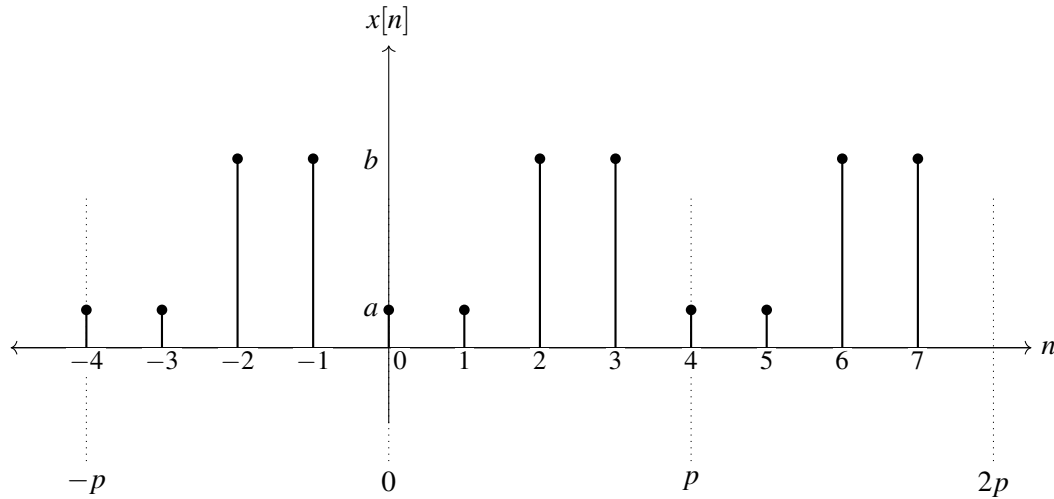
(c) Now suppose $y[n]$ is a p -periodic, real-valued, even signal, i.e. $y[n] = y[-n]$ for all $n \in \mathbb{Z}$. Let $\{Y_k \mid k \in \langle p \rangle\}$ be the DTFS coefficients of $y[n]$. Show that $Y_k \in \mathbb{R}$ for all $k \in \langle p \rangle$.

2. “Square” Waves

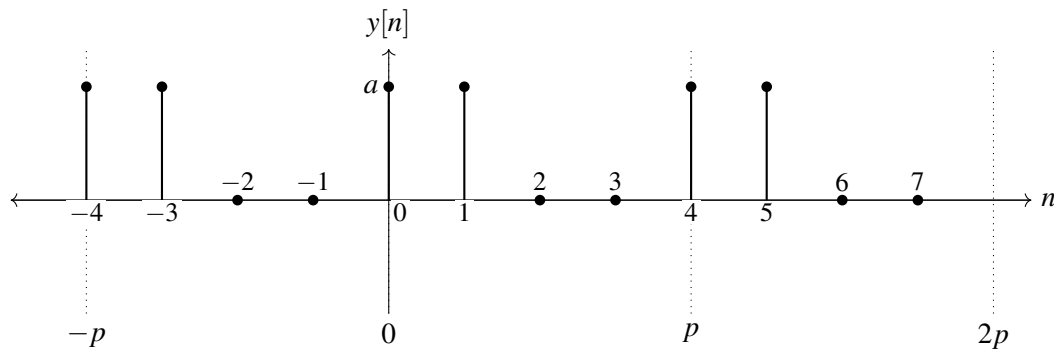
Consider a p -periodic, where p is even, “square” wave signal $x[n]$. It is defined over its first period as

$$x[n] = \begin{cases} a & \text{if } 0 \leq n < p/2 \\ b & \text{if } p/2 \leq n < p \end{cases}$$

And repeats periodically. To help you understand this notation, below is a plot of $x[n]$ for $p = 4$.

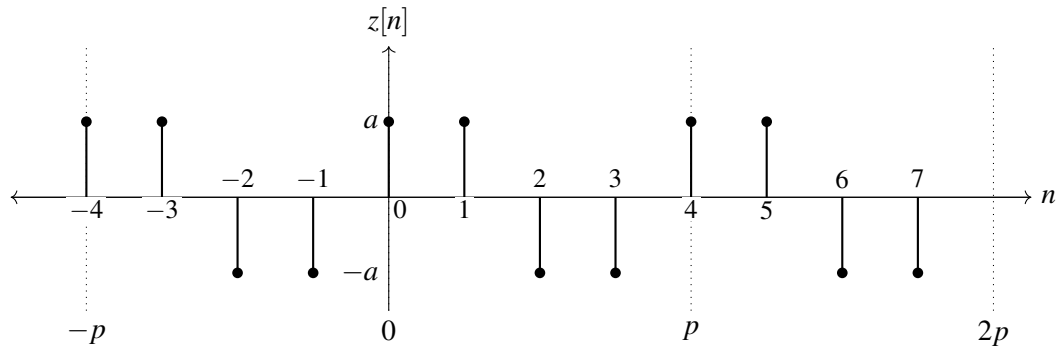


- Let X_k represent the DTFS coefficients of $x[n]$. Show that $X_0 = \frac{a+b}{2}$. You should prove this for a square wave with an arbitrary even period p . However, you do *not* need to compute *all* X_k for this problem.
- Now let consider a square wave with $b = 0$. Call this signal $y[n]$. For clarity, we plot this signal for $p = 4$:



Express the DTFS coefficient Y_k of $y[n]$ for general $k \in \langle p \rangle$. Your answer should be in terms of k , a , and p , and should include a summation.

- Now assume that $z[n]$ is a signal defined the same as $x[n]$, except the value of b is now equal to $-a$. This signal is now “half-wave symmetric.” Again, the case in which $p = 4$ is shown below:



Express the DTFS coefficients Z_k in terms of Y_k and k .

Hint: Recognize $z[n]$ as the sum of two square waves. Then, recall the time shift property: if $x[n]$ has DTFS coefficients X_k , then the shifted signal $x[n - n_0]$ has DTFS coefficients $e^{-ik\omega_0 n_0} X_k$.

- (d) Using your result from part (c), show that $Z_k = 0$ for any even $k \in \langle p \rangle$.

Hint: any even integer k can be expressed as $k = 2c$ for another integer c .

- (e) Another way to build $z[n]$ out of $y[n]$:

$$z[n] = 2y[n] - a.$$

Check that this visually makes sense, then write Z_k in terms of Y_k and compare with your answer to part (c). What does this tell you about Y_k ?

Hint 1: for a constant signal, the only non-zero DTFS coefficient is the $k = 0$ one.

3. Mystery DTFS (Adapted from EE 120 Discussion)

Consider a periodic real-valued discrete-time signal x having fundamental period p and an exponential discrete Fourier series expansion

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad (1)$$

where ω_0 is the fundamental frequency of the signal. The signal has the following properties:

- (i) $x[n+3] = x[n], \forall n \in \mathbb{Z}$
- (ii) $\sum_{n=0}^{p-1} x[n] = 0$
- (iii) $\sum_{k=0}^{p-1} X_k = 0$
- (iv) $\sum_{k=0}^{p-1} |X_k|^2 = \frac{1}{2}$

Show that the signal x can be expressed in the form $e^{i(An+B)} \sin(Cn)$, and determine what the parameters A , B , and C are. There may be multiple possible sets of parameters that satisfy the conditions.

Each property reveals something new about the DTFS expansion or the coefficients X_k ; consider them in the order given.

4. Parseval's Theorem

Let x be a **real valued** periodic signal that repeats every p samples. Additionally, let X_k be the coefficients in the DTFS expansion of x . It turns out that, when you transform into the DTFS basis, “energy” is conserved! This is stated formally by Parseval's theorem:

$$p \cdot \sum_{k=0}^{p-1} |X_k|^2 = \sum_{n=0}^{p-1} (x[n])^2$$

In this problem, we prove this fact using the linear-algebraic perspective of DTFS.

- (a) Let $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthogonal basis that spans $\mathbb{C}^n$, where each vector $\mathbf{v}_i$ has norm $\|\mathbf{v}_i\|$. In particular,

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} \|\mathbf{v}_i\|^2 & i = j, \\ 0 & i \neq j. \end{cases}$$

A vector $\mathbf{y} \in \mathbb{R}^n$ can be expressed in the V basis as

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \dots + \beta_n \mathbf{v}_n.$$

Write $\langle \mathbf{y}, \mathbf{y} \rangle$ in terms of $y_1, \dots, y_n$ **and** in terms of $\beta_1, \dots, \beta_n$.

Notice that even though $\mathbf{y}$ is real valued, the vectors $\mathbf{v}_i$ are **complex valued**. This is exactly what we do in DTFS! That is, we express a real valued signal in terms of the complex valued DTFS basis.

- (b) Using what we have learned above, show that:

$$p \cdot \sum_{k=0}^{p-1} |X_k|^2 = \sum_{n=0}^{p-1} (x[n])^2.$$

Hint: what is the norm of each DTFS basis vector?

5. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*
- (b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.
- (c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?
- (d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able to practice in this assignment?
- (e) Is there anything else you would like to share?